Geophys. J. Int. (1991) **105,** 397–405

Rapid rotation of southwest Japan—palaeomagnetism and K–Ar ages of Miocene volcanic rocks of southwest Japan

Yo-ichiro Otofuji,[1] Tetsumaru Itaya[2] and Takaaki Matsuda[3]

[1] *Department of Earth Sciences, Faculty of Science, Kobe University, Kobe* 657, *Japan*
[2] *Hiruzen Research Institute Okayama Branch, Okayama University of Science, Okayama* 700, *Japan*
[3] *Department of Geology, Himeji Institute of Technology, Himeji* 671-22, *Japan*

Accepted 1990 November 23. Received 1990 November 23; in original form 1990 May 18

SUMMARY

Miocene volcanic rocks have been sampled from the San'in district in the central part of southwest Japan. Reliable palaeomagnetic direction and K–Ar dating data have been obtained from 28 localities. A clockwise deflection in declination ($D = 38.5°$) is observed in the Kawai Formation having an age of 16.1 ± 1.4 Ma, whereas a northerly declination is observed in formations younger than 14.2 ± 0.6 Ma. These data indicate that southwest Japan had ceased rotational motion by about 14 Ma. When compared with the amount of rotation estimated from Cretaceous palaeomagnetic data, it is apparent that more than 80 per cent of the overall clockwise rotation occurred between about 16 and 14 Ma. We conclude that the angular velocity of southwest Japan about a rotation pole at 129°E, 34°N reached 20° Myr^{-1} at about 15 Ma. The corresponding rate of linear motion of the eastern part of southwest Japan was 21 cm yr^{-1}.

Key words: K–Ar dating, Japan Sea, palaeomagnetism, rotation, southwest Japan, tectonics.

INTRODUCTION

An important process at active continental margins is the rifting of island arcs associated with the opening of back-arc basins. Since asthenospheric material floods into the back-arc area during its opening (Garfunkel, Anderson & Schubert 1986), the motion of the island arc is controlled by flow of the asthenospheric material (Olson & Corcos 1980; Hager & O'Connell 1981; Schmeling & Jacoby 1981; Forte & Peltier 1987). Thus the dynamic process of motion of the island arc provides information on asthenospheric flow as well as the mechanical properties of the lithosphere beneath the arc–back-arc basin system.

It has become apparent in the last decade that southwest Japan experienced a large clockwise rotation associated with opening of the Japan Sea. Palaeomagnetic studies of the Japan Arc (Otofuji & Matsuda 1983, 1987; Otofuji, Hayashida & Torii 1985; Otofuji, Matsuda & Nohda 1985; Hayashida & Ito 1984; Hayashida 1986; Itoh 1988) and on-land field evidence (Murata 1987a, b; Faure, Fabbri & Monie 1988) indicate that southwest Japan rotated clockwise through about 45° with respect to the Asian continent in the Miocene, at about 15 Ma. The Japan Sea opened later than 19 Ma according to the evidence from the Japan Sea, including marine magnetic anomaly patterns (Isezaki 1975, 1986; Seama & Isezaki 1990), geochronological data (Kaneoka *et al.* 1990) and recent ODP Leg 127 drilling results (Leg 127 Scientific Drilling party 1989; Leg 127 & Leg 128 Shipboard Scientific Parties 1990). These data link the Miocene rotation of southwest Japan to the opening of the Japan Sea. The southwest Japan–Japan Sea system is, therefore, a particularly suitable objective for studying the dynamic processes of motion of an island arc associated with back-arc spreading.

The purpose of the present work is to elucidate the angular velocity of rotation of southwest Japan during these dynamic processes. We have investigated the palaeomagnetism and K–Ar ages of Miocene igneous rocks that outcrop in the San'in district of the central part of southwest Japan (Fig. 1). This has allowed the establishment of the rotation history of southwest Japan, from which the angular velocity can be estimated.

GEOLOGICAL SETTING AND SAMPLING

Extensive volcanic activity occurred in the Inner Zone of southwest Japan during the late Mesozoic and Tertiary periods (e.g. Murakami 1985). Rhyolitic to dacitic igneous products predominated in the early stages until the Early Oligocene, whereas dacitic to basaltic volcanism predominated in the Late Oligocene to Miocene.

The Iwami and Izumo Groups were formed during the

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

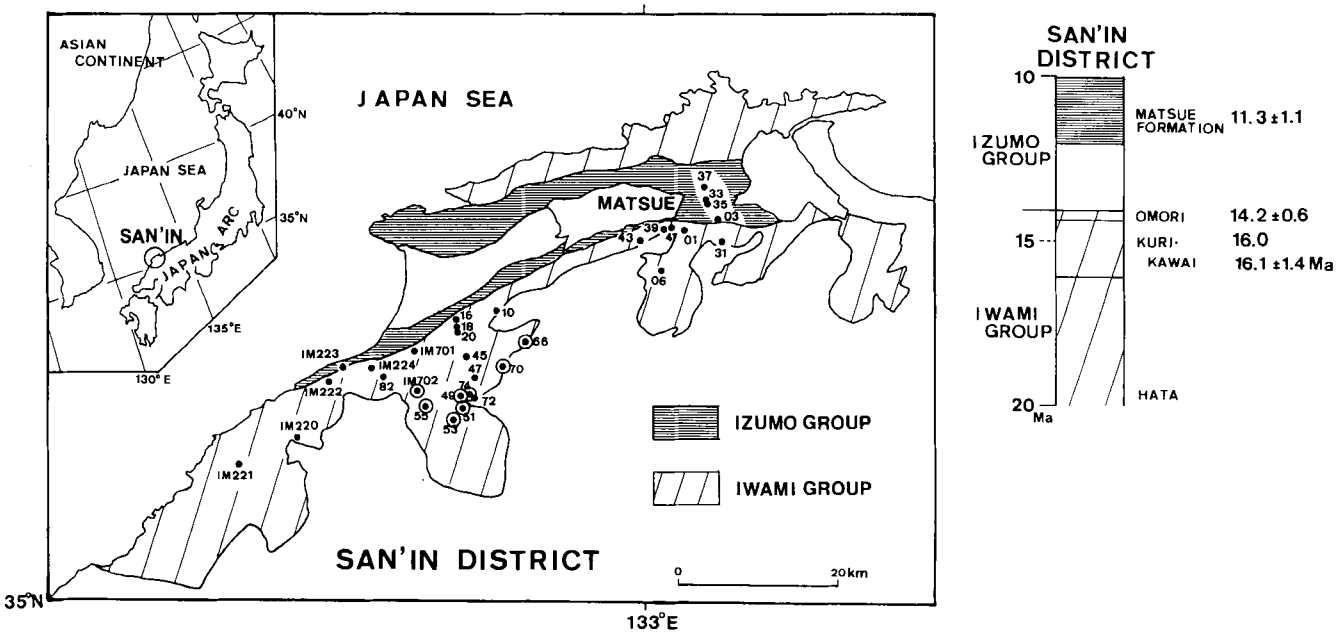


Figure 1. Generalized geologic map of sampling localities showing stratigraphic relationship in the San'in district of southwest Japan. Double circles mark sampling localities of the Kawai Formation and single solid circles those of the other formations. The K–Ar ages obtained in this study are assigned to the four formations. Map is adapted from Geological Map of Shimane Prefecture (Editorial Board of Geological Map of Shimane Prefecture 1982).

Miocene volcanic activity in the San'in district (Fig. 1). These groups are composed of the Hata, Kawai, Kuri, Omori and Matsue Formations in ascending order. Late Oligocene to Miocene ages (24 ~ 11.5 Ma) have been obtained for these groups (Matsuda 1983; Kano & Yoshida 1984; Kano & Nakano 1985). Sedimentary layers in the Kuri Formation dip gently to the north or northwestward at angles of less than 15°, so that the tilting of the younger volcanic rocks can be expected to be fairly gentle.

Our approach has been to measure both the palaeomagnetic direction and the K–Ar age of lava flows collected from each sampling site. Samples were collected at a total of 39 outcrops of the Kawai (12 sites), Kuri (one site), Omori (21 sites) and Matsue (four sites) Formations. A K–Ar sample was also collected from a site in the Omori Formation (IM 224) where reliable palaeomagnetic data had been previously obtained (Otofuji & Matsuda 1983). Hand samples for palaeomagnetic study were oriented by a magnetic compass. Each site typically comprised 12 independently oriented samples, distributed over a distance ranging up to 20 m.

K–Ar DATING

The K–Ar whole-rock dating was conducted following a petrographic study. Thin sections of the samples were examined to determine their suitability for whole-rock K–Ar dating, according to criteria discussed by Mankinen & Dalrymple (1972). 34 samples were found to be virtually free from alteration.

Whole-rock samples crushed to 80–100 mesh size were used for argon analysis. A portion of each sample was further ground in an agate mortar and used for potassium analysis. Details of the analytical procedures used for potassium and argon isotopic determinations have been reported by Itaya & Takasugi (1988). Potassium content was determined in duplicate by flame photometry using a 2000 ppm Cs buffer. The accuracy and reproducibility of the potassium analysis was within 2 per cent. Argon was analysed on a 15 cm radius sector type mass spectrometer with a single collector system at Okayama University of Science. The method of isotope dilution was employed, using an ^{38}Ar spike. Calibration of the ^{38}Ar spike is accurate to within about 1 per cent. Calculations of age and error followed the method described by Nagao & Itaya (1988) using the decay constants of Steiger & Jäger (1977).

The mean age and related uncertainty determinations for each formation are shown in Table 1. The mean age is computed by assigning unit weight to each age datum. The uncertainty Δt in age at the 95 per cent confidence level is expressed as $\Delta t = t_c S N^{-1/2}$, where N is the number of age determinations in each formation, S is the variance of the ages, and t_c is the confidence coefficient at the 97.5 per cent point on the Student's t distribution, with $N - 1$ degrees of freedom (e.g., Mandel 1984).

The ages of the Kawai Formation, the oldest volcanic rocks in our study, range between 13.7 and 18.3 Ma. The mean age and 95 per cent confidence limit are 16.1 Ma and ±1.4 Ma, respectively. The Kuri Formation yields one age datum of 16.0 Ma. A mean age from 15 samples of the Omori Formation is 14.2 Ma with a 95 per cent confidence limit of ±0.6 Ma. The uppermost Matsue Formation yields a mean age of 11.3 ± 1.1 Ma.

PALAEOMAGNETISM

A palaeomagnetic study was conducted on samples from the 33 sampling sites on which the K–Ar age determinations were made. In the laboratory, individual specimens, 25 mm in diameter and 25 mm long, were prepared from the

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

Table 1. Palaeomagnetic directions and K–Ar dating results from Miocene volcanic rocks of San'in district. Palaeomagnetic directions without tilt correction are presented. Rock type: a = andesite; b = basalt; d = dacite; Do = dolerite. N = number of samples, DEC = declination; INC = inclination; DEMAG.: A = single-component magnetization; B = two-component magnetization. The optimum demagnetization temperature (ODT; see text) or temperature range used to calculate the NRM direction by principal component analysis is shown in °C. ORG = origin. The magnetization directions defined by both demagnetization at an ODT and by principal component analysis (Kirschvink 1980) are displayed for the two sites OM41 and OM66. The magnetization direction of the low-temperature component is listed for site OM70. K = precision parameter, α_{95} = radius of 95 per cent confidence about the mean direction. Palaeomagnetic data denoted by asterisk are from Otofuji & Matsuda (1983). All of the age data are original. Ages are calculated from the following decay and abundance constants: $\lambda_\varepsilon = 0.581 \times 10^{-10}\,\mathrm{yr}^{-1}$; $\lambda_\beta = 4.962 \times 10^{-10}\,\mathrm{yr}^{-1}$; $^{40}\mathrm{K/K} = 1.167 \times 10^{-4}\,\mathrm{mol\ mol}^{-1}$ (Steiger & Jäger 1977). The mean age and its 95 per cent confidence limit are shown for each formation.

FORMATION	SITE	LAT	LON	ROCK TYPE	N	DEC(°)	INC(°)	DEMAG.	K	α_{95}(°)	K-Ar AGE (Ma)	Potassium (wt. %)	Rad. argon 40 (10^{-8}ccSTP/g)	Air conc. (%)
Matsue	OM03	35°26.1'	133°06.3'	b	12	-149.6	-57.0	A 300	319.0	2.4	10.8±0.2	1.26±0.03	53.0±0.4	27.1
	OM33	35°27.1'	133°05.1'	b	12	-5.7	55.9	A 400	23.8	8.7	(11.5±0.3)			
	OM35	35°27.1'	133°05.1'	b	8	-10.5	54.4	A 400	189.0	4.0	11.5±0.3	2.64±0.05	118.0±1.9	33.7
	OM37	35°28.2'	133°05.1'	b	10	172.1	-25.8	B 200~ORG	19.2	11.3	11.6±0.3	2.44±0.05	110.2±1.2	11.7
	mean										11.3±1.1(N=3)			
Omori	IM221*	35°27.2'	132°27.2'	a	8	-165.3	-58.7	500	41.7	8.7				
	IM222*	35°15.0'	132°34.6'	a	12	164.7	-39.6	400	210.3	3.0				
	IM223*	35°16.1'	132°35.7'	d	12	148.5	-65.9	400	242.9	2.8				
	IM224*	35°16.1'	132°38.0'	a	12	-3.0	57.2	400	7.8	16.6	15.7±0.6	0.61±0.02	37.4±0.8	64.3
	IM701	35°17.1'	132°41.5'	a	10	71.0	56.8	A 400	195.4	3.5	13.0±0.5	0.73±0.02	37.0±0.7	64.5
	OM01	35°25.3'	133°03.3'	a	9	5.4	54.0	A 300	146.4	4.3	14.4±0.3	1.81±0.04	101.6±0.7	11.6
	OM06	35°22.7'	133°01.5'	a	12	40.2	72.4	A 300	174.0	3.3	14.1±0.3	1.78±0.04	97.5±0.6	5.8
	OM10	35°20.0'	132°48.0'	b	12	-22.4	47.3	A 300	111.2	4.1	16.2±1.4	0.29±0.02	18.3±1.4	89.0
	OM16	35°19.2'	132°44.8'	a	12	-10.5	35.2	A 300	48.0	6.3	14.4±0.3	1.50±0.03	84.2±0.5	12.1
	OM18	35°18.2'	132°44.9'	b	12	7.7	21.9	A 300	68.7	5.3	13.4±0.3	1.07±0.02	55.7±0.8	55.4
	OM20	35°18.2'	132°44.9'	a	11	-18.6	22.6	A 300	547.8	2.0	13.8±0.3	1.60±0.03	85.8±1.0	47.0
	OM39	35°25.5'	133°01.7'	a	11	2.4	62.6	A 400	171.0	3.5	13.0±0.3	1.94±0.04	98.2±1.4	24.5
	OM41	35°25.5'	133°02.3'	a	11	-9.1	65.1	A 450	164.2	3.6	13.2±0.3	2.07±0.04	106.6±1.2	15.1
					(11	-8.7	64.8	100~ORG	129.9	4.0)				
	OM43	35°24.8'	132°59.7'	a	11	28.4	77.2	B 350~ORG	53.6	6.3	12.6±0.3	1.30±0.03	63.8±0.7	14.1
	OM45	35°16.5'	132°45.8'	d	11	14.1	51.7	A 500	12.5	12.8	15.3±0.4	2.08±0.04	124.2±1.4	15.6
	OM47	35°15.2'	132°46.4'	a	12	-0.2	51.8	A 300	32.5	7.7	13.9±0.6	0.95±0.03	51.1±1.3	54.7
	OM72	35°13.8'	132°46.3'	Do	11	170.2	-54.7	B 400~ORG	53.4	6.3	14.9±0.9	0.30±0.02	17.1±0.7	66.1
	OM82	35°15.4'	132°39.0'	a	11	3.3	73.7	B 300~ORG	28.9	8.6	15.3±0.6	0.89±0.03	53.1±1.2	50.7
	mean										14.2±0.6(N=15)			
Kuri	IM220*	35°11.0'	132°32.0'	d	12	25.0	50.9	500	14.0	12.0				
	OM31	35°24.6'	133°06.5'	d	11	-48.1	-35.3	B 300~ORG	28.4	8.7	16.0±0.4	1.83±0.04	114.3±1.8	33.7
Kawai	IM702	35°14.4'	132°41.7'	a	11	32.8	46.9	B 350~ORG	107.4	4.4	18.3±0.4	1.16±0.02	82.7±0.6	24.4
	OM49	35°14.1'	132°45.2'	a	11	53.1	51.3	B 350~ORG	131.3	4.0	16.2±0.4	1.26±0.03	79.2±1.0	19.9
	OM51	35°13.4'	132°45.3'	a	13	36.4	35.0	B 300~ORG	51.3	5.8	15.2±0.4	1.10±0.02	64.8±0.9	26.6
	OM53	35°12.5'	132°44.8'	a	7	43.2	44.9	B 300~ORG	30.5	11.1	17.5±1.4	0.19±0.01	13.1±0.8	77.7
	OM55	35°13.3'	132°42.4'	a	11	28.7	58.1	B 400~ORG	155.6	3.7	13.8±0.4	1.20±0.02	64.4±1.2	42.7
	OM66	35°17.7'	132°50.5'	b	9	5.0	35.5	A 300	28.3	9.8	17.3±1.3	0.20±0.01	13.3±0.8	77.5
					(9	2.2	37.1	100~ORG	35.2	8.8)				
	OM70	35°16.1'	132°48.6'	a	11	65.0	6.3	B 200~ORG	12.1	13.7	16.4±0.4	1.38±0.03	88.1±1.1	19.8
					(6	5.8	43.4	0~300	19.9	15.4)				
	OM74	35°13.9'	132°46.1'	a	11	36.7	29.1	B 450~ORG	156.3	3.7	13.7±0.3	1.04±0.02	55.5±0.9	32.5
	mean										16.1±1.4(N=8)			

samples. Natural remanent magnetizations were measured with either a spinner magnetometer (Schonstedt SSM-1A or a laboratory made magnetometer) or a higher sensitivity ScT cryogenic magnetometer according to the intensity of magnetization.

Stepwise thermal demagnetization was carried out for three pilot specimens from each site in steps of 30° to 50 °C; preliminary demagnetization experiments indicated that thermal demagnetization was effective for erasing low blocking temperature components of magnetization. Thermal demagnetization was conducted using a laboratory built furnace. The residual magnetic field in which the specimens were cooled was kept less than 10 nT. Zijderveld component-plots (Zijderveld 1967) and equal area projections were used to assess the directional stability and unblocking temperature spectrum for each specimen.

Results from 27 sites out of 33 showed that the pilot specimens exhibited similar demagnetization behaviour and their stable magnetization directions were clustered. The other sites showed a large variety in demagnetization behaviour among the pilot specimens: their magnetization directions at the demagnetization end points were scattered. Results from these six sites were excluded from the final analysis.

12 sites out of these 27 exhibited two distinct magnetization components (Fig. 2). The characteristic direction of the high-temperature component was determined using principal component analysis (Kirschvink 1980). These vectors were then combined using Fisher (1953) statistics. The unblocking temperature of the high-temperature component ranged between 560° and 600 °C, indicating that this component is carried by titanium-poor magnetite. The low-temperature components were removed in the initial stage of demagnetization up to 350 °C. Their directions are dominated by a component in the present direction of the Earth's magnetic field. The low-temperature component is probably a viscous partial thermoremanent magnetization acquired in the recent geomagnetic field (Pullaiah *et al.* 1975; Kent 1985).

The other 15 sites exhibited a single magnetic component,

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

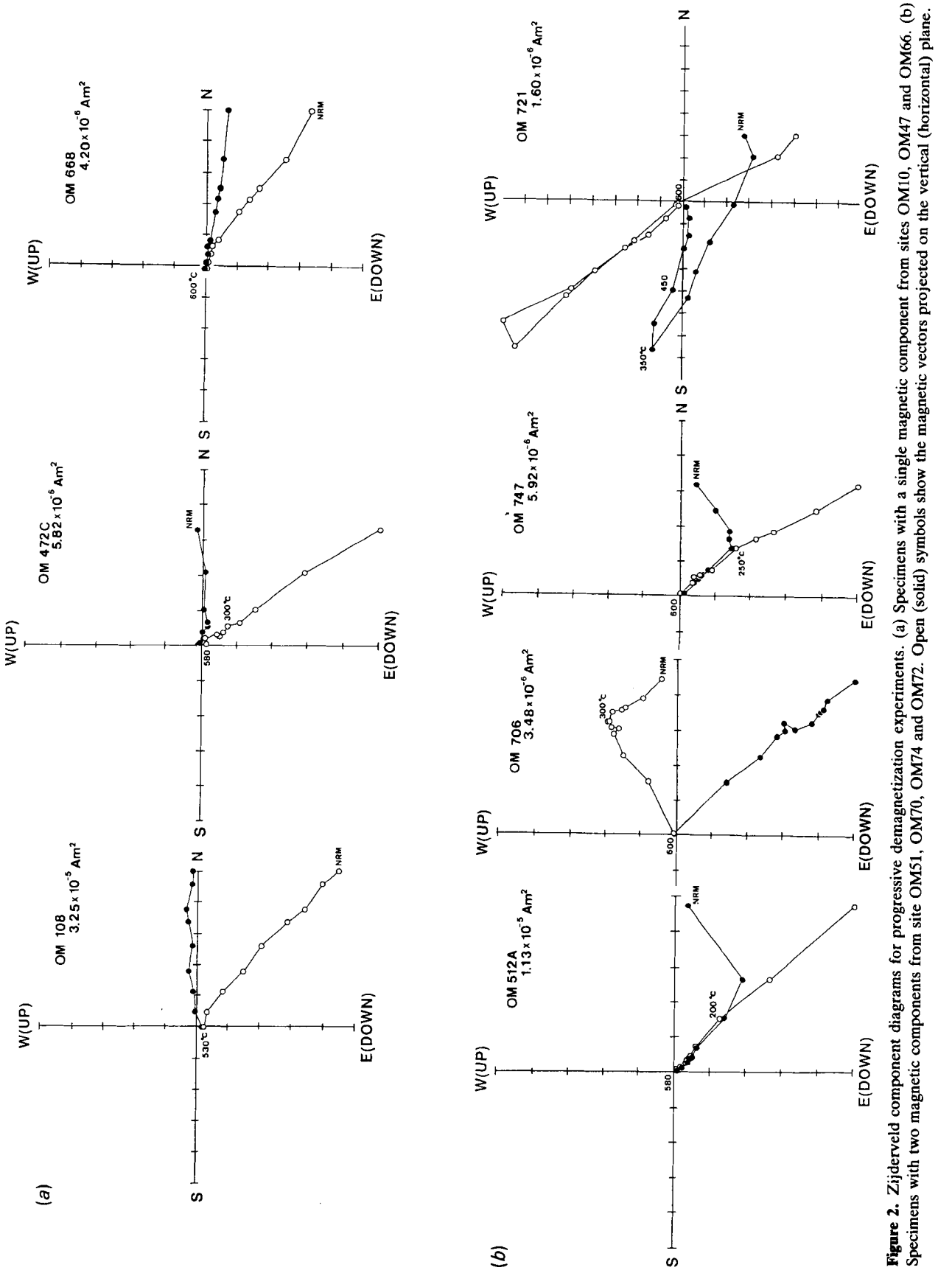


Figure 2. Zijderveld component diagrams for progressive demagnetization experiments. (a) Specimens with a single magnetic component from sites OM10, OM47 and OM66. (b) Specimens with two magnetic components from site OM51, OM70, OM74 and OM72. Open (solid) symbols show the magnetic vectors projected on the vertical (horizontal) plane.

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

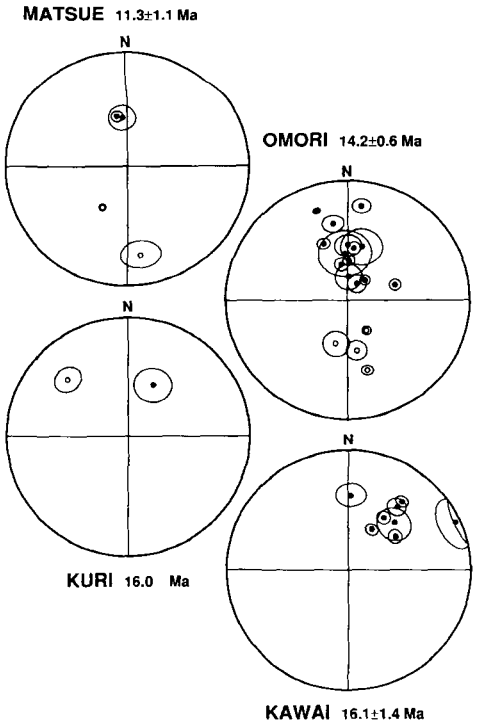


Figure 3. The mean characteristic magnetization directions and 95 per cent confidence circles for sites in the Matsue, Omori, Kuri and Kawai Formations. The mean K–Ar ages are shown for each formation. Projections are equal area, solid (open) symbols on the lower (upper) hemisphere.

which was unblocked at about 550° ~ 600 °C (Fig. 2). Remaining specimens from these sites were demagnetized at the optimum demagnetization temperature (ODT), i.e. that which produced minimum dispersion of the magnetization directions for the three pilot specimens. The ODT values range between 300° and 500 °C, at which temperature the viscous partial thermoremanent magnetization of magnetite is probably removed (Pullaiah *et al.* 1975; Kent 1985). Similar magnetization directions were obtained by both principal component analysis (Kirschvink 1980) and demagnetization at an ODT, as illustrated for the two sites OM41 and OM66 (see Table 1).

Palaeomagnetic data are presented in Fig. 3 and Table 1 together with previously published data from the same formations (Otofuji & Matsuda 1983, 1984). The circles of 95 per cent confidence do not exceed 17° for any of the 32 sites. Site mean directions were combined to obtain the mean direction for each formation listed in Table 2. The mean palaeomagnetic directions in Table 2 refer to the value at a representative point (134°E, 35°N) in southwest Japan.

Kawai Formation

Samples from all sites carry only a normal polarity magnetization. The magnetizations of seven of the eight sites show two distinct components (Fig. 2b). Characteristic high-temperature component directions show a significant clockwise deflection. Declinations range from 29° to 65°. The mean direction is $D = 44°$, $I = 40°$. The mean inclination value is lower than that for the present axial dipole field at the sampling localities ($I = 54°$). Only one site, OM66, has a single component magnetization (Fig. 2a) and its palaeomagnetic direction shows a northerly declination ($D = 5°$, $I = 36°$, $\alpha_{95} = 10°$). A similar direction is obtained by principal component analysis ($D = 2°$, $I = 37°$, $\alpha_{95} = 9°$). Although this declination value is close to that of the present geomagnetic field direction, the inclination value corresponds to the low value observed for the other sites in the Kawai Formation. The northerly declination may possibly result from secular variation in Miocene time. The overall mean direction ($D = 39°$, $I = 41°$, $\alpha_{95} = 15°$) which includes the results from OM66, is believed to best represent the characteristic palaeomagnetic direction of the Kawai Formation.

Kuri Formation

Only one site was sampled in this formation. This palaeomagnetic direction is quite anomalous, showing a westerly declination and upward inclination ($D = -48°$, $I = -35°$). This direction is inconsistent with the easterly declination which was observed in the previous study of site IM 220 in this formation (Otofuji & Matsuda 1983) (see Table 1).

Omori and Matsue Formations

Northerly and southerly declinations are observed in these formations, representing normal and reversed polarity characteristic magnetizations respectively. The inclination values are quite steep ($I_{Omori} = 56°$, $I_{Matsue} = 49°$) and similar to that of the present axial dipole field at the sampling sites.

DISCUSSION

Reliability of K–Ar dating and palaeomagnetism

K–Ar age data are presented in stratigraphic sequence in Fig. 1. This illustrates the good agreement of the K–Ar data with observed stratigraphic relationships.

Table 2. Average palaeomagnetic directions and pole positions for each of the Miocene formations studied. Directions are presented as the value at the representative point of southwest Japan (134°E, 35°N). N is number of sites, D, I are mean declinations and inclinations, and α_{95} and k are Fisher statistical parameters.

Formation	age (Ma)	N	D(°)	I(°)	α95(°)	pole position long.	lat.	α95(°)
Matsue	11.3 ± 1.1	4	-0.2	48.9	22.4	55.6°W	86.2°N	22.9
Omori	14.2 ± 0.6	18	-0.1	55.6	9.0	146.0°W	86.1°N	10.5
Kawai	16.1 ± 1.4	8	38.5	40.7	15.1	128.5°W	55.7°N	14.2

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

The K–Ar ages are in good agreement with the palaeontologically assigned ages of Early/Middle Miocene. Huang & Okamoto (1979) show, on the basis of the planktonic foraminifera and calcareous nannofossils present in interbedded sediments, that the Kawai Formation belongs to the N8 ~ N10 planktonic foraminiferal zone of Blow (1969). Tai & Kato (1979) suggest that the lower part of the Kuri Formation belongs to Blow's Zone N9. The numerical ages for the lowest part of the Zone N9 and N8 are estimated at 15.1 and 16.1 Ma, respectively (Hsü *et al.* 1984). These ages are close to our K–Ar ages of 16.0 Ma for the Kuri Formation and 16.1 Ma for the Kawai Formation.

The most striking aspect of the palaeomagnetic results is the relatively large clockwise deflection of the declination of the Kawai Formation. The reliability of the easterly declination is demonstrated through the resistance of this component to thermal demagnetization.

Palaeomagnetic data obtained from this study are consistent with previously established palaeomagnetic results from this area. Virtual geomagnetic pole (VGP) positions for the Kawai and Omori Formations are shown in Fig. 4, together with the apparent polar wander path (APWP) for the San'in district for the interval 100 to 20 Ma (Otofuji & Matsuda 1987). The VGP for the Kawai Formation is located at 129°W, 56°N; This lies between the pole for 20 Ma and the geographic north pole. The VGP for the Omori Formation is located close to the geographic north pole. The VGPs for the Miocene formations form a consistent chronological progression. We conclude that our chronological and palaeomagnetic data reliably record rotational motion for southwest Japan.

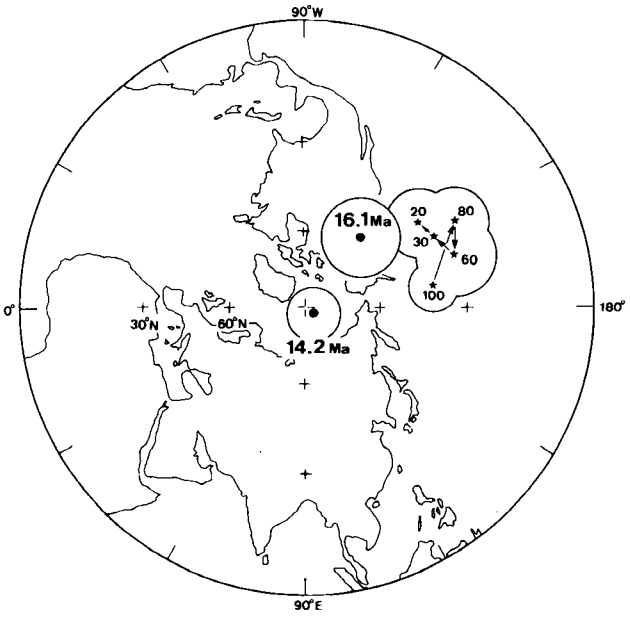


Figure 4. Pole positions with 95 per cent circles of confidence for the Omori and Kawai Formations in the San'in district, compared with the apparent polar wander path (star) for the interval 100 to 20 Ma for southwest Japan (Otofuji & Matsuda 1987). Poles for the Omori and Kawai Formations are labelled by 14.2 Ma and 16.1 Ma, respectively. Polar stereographic projection.

Rotation of southwest Japan

The mean declinations for the Kawai, Omori and Matsue Formations are taken to represent the amount of rotation of southwest Japan with respect to the eastern part of Asia because the VGP for China, younger than 25 Ma is located close to the geographic north pole (Zhao *et al.* 1990; Zheng *et al.* 1990). Available rotation data for southwest Japan are plotted as a function of age for the interval 100 ~ 10 Ma in Fig. 5.

The plot shows that most of the rotational motion (~40°) occurred between 16.1 and 14.2 Ma. Rotational motion appears to have virtually ceased by 14.2 Ma. Only a few degrees (5° to 10°) of rotation can be accounted for between 16.1 Ma and the Cretaceous Period. A total rotation of ~44°–49° since the Cretaceous Period was suggested by Otofuji & Matsuda (1987). The present study shows that more than 80 per cent of this total clockwise rotation of southwest Japan occurred later than about 16 Ma.

This estimate of the age for the bulk of the rotation is consistent with the timing for the climax of the rotational motion at 14.9 Ma, which is estimated from palaeomagnetic studies of Miocene sedimentary rocks (Hayashida 1986) and from a compilation of Neogene palaeomagnetic data from

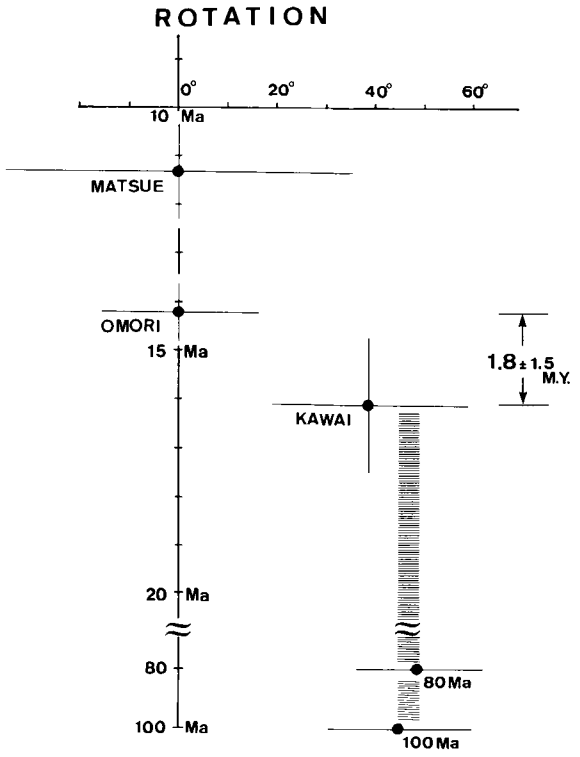


Figure 5. Rotation with respect to the eastern part of Asia versus age for geological units in the San'in district of southwest Japan. The rotation error bars are derived from $\Delta R = \sin^{-1}$ [sin $\alpha_{95}/\cos(I)$] where I and α_{95} are the inclination and the radius of 95 per cent confidence about the mean direction. The age error bars are the 95 per cent confidence limits. The shaded rotation zone is the amount of clockwise rotation of southwest Japan with respect to the eastern part of Asia estimated on the basis of the Cretaceous palaeomagnetic data (Otofuji & Matsuda 1987).

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

southwest Japan (Otofuji *et al.* 1985a). The present study provides independent evidence for the timing of rotation of southwest Japan.

It appears that a steepening of the magnetic inclination by about 15° is associated with the rotational motion between 16 and 14 Ma ($I_{Omori} = 56°$ versus $I_{Kawai} = 41°$). However, the uncertainty of this value at the 95 per cent confidence level $\{\Delta I = [(\alpha_{95\ Omori})^2 + (\alpha_{95\ Kawai})^2]^{1/2}\}$ is ±18°. Thus the difference in inclination values is 15° ± 18°, which is not statistically significant. Further studies are required to establish the significance of the apparent inclination changes during Miocene time.

Angular velocity and drift velocity of southwest Japan

Our data indicate that a fast rotation of southwest Japan occurred in the Miocene. Accounting for the dispersion in the data for the Kawai Formation, the amount of rotation (R) and its 95 per cent confidence level [$\Delta R = \sin^{-1}(\sin \alpha_{95}/\cos I)$] is 38.5° ± 20.1°. An estimate of the maximum duration of the rotation period is given by $T = t_{Kawai} - t_{Omori} = 16.05 - 14.21 = 1.84$ Myr, and the corresponding 95 per cent confidence limit is $\Delta T = (\Delta t^2_{Kawai} + \Delta t^2_{Omori})^{1/2} = 1.54$ Myr, where Δt is the uncertainty in age at the 95 per cent confidence limit for each formation. The amount of rotation and the duration of the rotation period, together yield a mean angular velocity (μ) of 38.5°/1.84 = 21° Myr^{-1} about a rotation pole located at *Long* = 134°E, *Lat* = 35°N. The 95 per cent confidence interval ($\Delta\mu$) for the mean velocity is 20° Myr^{-1}; $\Delta\mu = \mu[(\Delta R/R)^2 + (\Delta T/T)^2]^{1/2}$. Although this 95 per cent confidence interval is quite large, our preferred interpretation is that southwest Japan rotated at an average rate of 21° Myr^{-1} at about 15 Ma.

The drift velocity of southwest Japan is evaluated on the basis of a rigid block rotation model. Detailed Tertiary palaeomagnetic results (Sasajima, Nishida & Shimada 1968; Nakajima & Hirooka 1986; Itoh 1988; Itoh & Ito 1989; Nakajima *et al.* 1990) indicate that southwest Japan rotated as a rigid unit; the unit forms a block extending from Kyushu to the west of the Nohi area, and is about 600 km long. This rigid block was rotated about a pole close to the western end of the area, based on geological and geomorphological constraints (Murata 1987a,b; Faure *et al.* 1988; Honza 1979). We locate the pole at 129°E, 34°N after Otofuji & Matsuda (1987) (Fig. 6). The mean angular velocity about this pole is 37.4°/1.84 = 20° Myr^{-1} i.e. when southwest Japan is rotated about this pole through 37.4°, the declination changes from 38.5° to 0° (Perroud 1982; Matsuda, Matsubara & Nishimura 1985). These results imply that the eastern part of southwest Japan moved at a velocity of $2\pi \times 600$ km $\times$ 20°/360° = 209 km Myr^{-1} = 21 cm yr^{-1} (Fig. 6).

These results, combined with our knowledge of the opening of the Japan Sea, suggest that the spreading took place at a fast rate of 21 cm yr^{-1} in the central part of the Japan Sea at around 15 Ma. The fast spreading would be associated with the inflow of the asthenosphere with low viscosity, which is observed beneath the Japan arc (Rydelek & Sacks 1988). Several models have been proposed for the opening of the Japan Sea (e.g. Lellemand & Jolivet 1985; Celaya & McCabe 1987). Our estimate of the timing and fast spreading rate will allow refinement of these models.

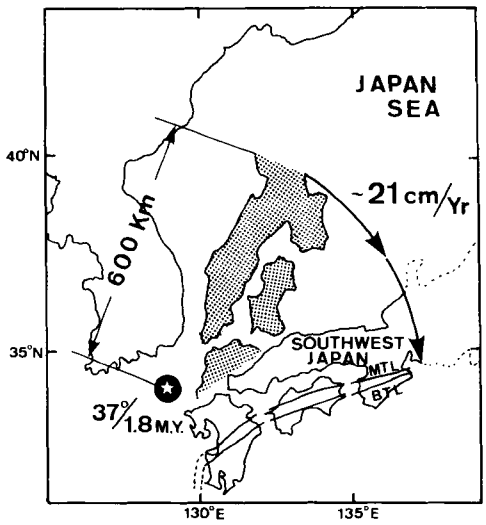


Figure 6. Drifting velocity of the eastern part of southwest Japan. Southwest Japan rotated about the rotation pole at 129°E, 34°N with an angular velocity of about 20° Myr^{-1} for about 1.8 Myr at about 15 Ma. *The eastern part of southwest Japan, around 600 km* from the rotation pole, drifted with a velocity of about 21 cm yr^{-1}. MTL: Median Tectonic Line. BTL: Butsuzo Tectonic Line.

ACKNOWLEDGMENTS

We are indebted to Professor S. Nohda for illuminating conversation on the opening of the Japan Sea. The assistance of Messrs M. Kamino and A. Ishitani in the laboratory work is appreciated. Anonymous referees provided useful suggestions and helped to improve the English. Measurements of remanent magnetization with a Schonstedt SSM-1A spinner magnetometer were made at the palaeomagnetic laboratory in the Department of Geology and Mineralogy of Kyoto University, by courtesy of Professor M. Torii. This work was supported in part by the Asahi scholastic promotion fund, the Itoh Science Foundation and a Grant-in-aid (No. 01540633) from the Japanese Ministry of Education, Science and Culture.

REFERENCES

Blow, W. H., 1969. Late Middle Eocene to Recent planktonic foraminiferal biostratigraphy, in *Proc. 1st Int. Conf. Planktonic Fossils*, Geneva, pp. 199–422, E. J. Brill, Leiden.

Celaya, M. & McCabe, R., 1987. Kinematic model for the opening of the Sea of Japan and the bending of the Japanese islands, *Geology*, **15**, 53–57.

Editorial Board of Geological Map of Shimane Prefecture, 1982. *Geological Map of Shimane Prefecture*, Department of Geology, Shimane University.

Faure, M., Fabbri, O. & Monie, P., 1988. The Miocene bending of Southwest Japan: new $^{39}Ar/^{40}Ar$ and microtectonic constraints from Nagasaki schists (western Kyushu), an extension of the Sanbagawa high-pressure belt, *Earth planet. Sci. Lett.*, **91**, 105–116.

Fisher, R. A., 1953, Dispersion on a sphere, *Proc. R. Soc. Lond.*, A., **217**, 295–305.

Forte, A. M. & Peltier, W., 1987. Plate tectonics and aspherical

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

structure: The importance of poloidal–toroidal coupling, *J. geophys. Res.*, **92,** 3645–3679.

Garfunkel, Z., Anderson, C. A. & Schubert, G., 1986. Mantle circulation and the lateral migration of subducted slabs, *J. geophys. Res.*, **91,** 7205–7223.

Hager, B. H. & O'Connell, R. J., 1981. A simple global model of plate dynamics and mantle convection, *J. geophys. Res.*, **86,** 4843–4867.

Hayashida, A., 1986. Timing of rotational motion of Southwest Japan inferred from paleomagnetism of the Setouchi Miocene Series, *J. Geomag. Geoelectr.*, **38,** 295–310.

Hayashida, A & Ito, Y., 1984. Paleoposition of southwest Japan at 16 Ma: implication from paleomagnetism of the Miocene Ichishi Group, *Earth planet. Sci. Lett.*, **68,** 335–342.

Honza, E., 1979. Sediments, structure, and origin of Japan Sea—concluding remarks, *Geol. Surv. Japan, Cruise Rep.*, **13,** 89–93.

Hsü, K. J., Brecque, J. La. *et al.*, 1984. Numerical ages of Cenozoic biostratigraphic datum levels: results of South Atlantic Leg 73 drilling, *Geol. Soc. Am. Bull.*, **95,** 863–876.

Huang, T. & Okamoto, K., 1979. Planktonic Foraminiferal assemblages from the Kawai Formation and its correlatives in the San-in Area, Japan, *Bull. Mizunami Fossil Mus.*, **6,** 101–110.

Isezaki, N., 1975. Possible spreading centers in the Japan Sea, *Mar. geophys. Res.*, **2,** 265–277.

Isezaki, N., 1986. A magnetic anomaly map of the Japan Sea, *J. Geomag. Geoelectr.*, **38,** 403–410.

Itaya, T. & Takasugi, H., 1988. Muscovite K–Ar ages of the Sanbagawa schists, Japan and argon depletion during cooling and deformation, *Contrib. Mineral. Petrol.*, **100,** 281–290.

Itoh, Y., 1988. Differential rotation of the eastern part of southwest Japan inferred from paleomagnetism of Cretaceous and Neogene rocks, *J. geophys. Res.*, **93,** 3401–3411.

Itoh, Y. & Ito, Y., 1989. Confined ductile deformation in the Japan arc inferred from paleomagnetic studies, *Tectonophysics*, **167,** 57–73.

Kaneoka, I., Notsu, K., Takigami, Y., Fujioka, K. & Sakai, H., 1990. Constraints on the evolution of the Japan Sea based on ^{40}Ar–^{39}Ar ages and Sr isotopic ratios for volcanic rocks of the Yamato Seamount chain in the Japan Sea, *Earth planet. Sci. Lett.*, **97,** 211–225.

Kano, K. & Yoshida, F., 1984. Radiometric ages of the Neogene in central eastern Shimane prefecture, and their implication in stratigraphic correlation, *Bull. geol. Surv. Japan*, **35,** 159–170.

Kano, K. & Nakano, S., 1985. Stratigraphic correlation of the Neogene in the San-in district, Southwest Japan, in the light of radiometric dating, *Bull. geol. Surv. Japan*, **36,** 427–438.

Kent, D. V., 1985. Thermoviscous remagnetization in some Appalachian limestones, *Geophys. Res. Lett.*, **12,** 805–808.

Kirschvink, J. L., 1980. The least squares line and plane and the analysis of palaeomagnetic data, *Geophys, J. R. astr. Soc.*, **62,** 699–718.

Leg 127 Scientific Drilling Party, 1989. Exploring the Japan Sea, *Geotimes*, **34,** no. 12, 19–21.

Leg 127 & Leg 128 shipboard scientific party, 1990. Evolution of the Japan Sea, *Nature*, **346,** 18–20.

Lellemand, S. & Jolivet, L., 1985. Japan Sea: a pull-apart basin, *Earth planet. Sci. Lett.*, **76,** 375–389.

Mandel, J., 1984. *The Statistical Analysis of Experimental Data*, Dover, New York.

Mankinen, E. A. & Dalrymple, G. B., 1972. Electron microscope evaluation of terrestrial basalts for whole-rock K–Ar dating, *Earth planet. Sci. Lett.*, **17,** 89–94.

Matsuda, T., 1983. Fission-track geochronology and tectonics of late Cretaceous to early Miocene igneous rocks in central to west San'in, Southwest Japan, *Doctoral thesis*, Faculty of Science, Kyoto University.

Matsuda, J., Matsubara, Y. & Nishimura, J., 1985. The determination of 'paleolatitude' and its geophysical application to the Bonin Islands, *J. Geomag. Geoelectr.*, **37,** 561–574.

Murakami, N., 1985. Late Mesozoic to Paleogene igneous activity in West Chugoku, Southwest Japan, *J. geol. Soc. Japan*, **91,** 723–742.

Murata, A., 1987a. Hokusatsu bend and clockwise rotation of the Southwest Japan Arc, *J. Fac. Sci. Univ. Tokyo*, **21,** 333–349.

Murata, A., 1987b. Conical folds in the Hitoyoshi Bending, South Kyushu, formed by the clockwise rotation of the Southwest Japan Arc, *J. geol. Soc. Japan*, **93,** 91–105.

Nagao, K. & Itaya, T., 1988. K–Ar age determination, *Mem. geol. Soc. Japan*, **29,** 5–21.

Nakajima, T. & Hirooka, K., 1986. Clockwise rotation of Southwest Japan inferred from paleomagnetism of Miocene rocks in Fukui prefecture, *J. Geomag. Geoelectr.*, **38,** 513–522.

Nakajima, T., Sawada, Y., Nakagawa, T., Hayashi, A. & Itaya, T., 1990. Paleomagnetic results and K–Ar dating on Miocene rocks in the northern part of Fukui Prefecture, Central Japan—with reference to the rotation of Southwest Japan, *J. Jpn Assoc. Mineral. Petrol. Econ. Geol.*, **85,** 45–59.

Olson, P. & Corcos, G. M., 1980. A boundary layer model for mantle convection with surface plates, *Geophys. J. R. astr. Soc.*, **62,** 195–219.

Otofuji, Y. & Matsuda, T., 1983. Paleomagnetic evidence for the clockwise rotation of Southwest Japan, *Earth planet. Sci. Lett.*, **62,** 349–359.

Otofuji, Y. & Matsuda, T., 1984. Timing of rotational motion of Southwest Japan inferred from paleomagnetism, *Earth planet. Sci. Lett.*, **70,** 373–382.

Otofuji, Y. & Matsuda, T., 1987. Amount of clockwise rotation of Southwest Japan—fan shape opening of the southwestern part of the Japan Sea, *Earth planet. Sci. Lett.*, **85,** 289–301.

Otofuji, Y., Hayashida, A. & Torii, M., 1985. When was the Japan Sea opened?: Paleomagnetic evidence for Southwest Japan, in *Formation of Active Margins*, pp. 551–566, eds Nasu, N., Uyeda, S., Kushiro, I., Kobayashi, K. & Kagami, H., Terrapub., Tokyo.

Otofuji, Y., Matsuda, T. & Nohda, S., 1985. Opening mode of the Japan Sea inferred from paleomagnetism of the Japan arc, *Nature*, **317,** 603–604.

Perroud, H., 1982. The change of palaeomagnetic vector orientation induced by Eulerian rotations: applications for the relative rotations of Spain and Europe, *Tectonophysics*, **81,** T15–T23.

Pullaiah, G. E., Irving, E., Buchan, K. L. & Dunlop, D. J., 1975. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.*, **28,** 133–143.

Rydelek, P. A. & Sacks, I. S., 1988. Asthenospheric viscosity inferred from correlated land–sea earthquakes in north-east Japan, *Nature*, **336,** 234–237.

Sasajima, S., Nishida, J. & Shimada, M., 1968. Paleomagnetic evidence of a drift of the Japanese Main Island during the Paleogene period, *Earth planet. Sci. Lett.*, **5,** 135–141.

Schmeling, H. & Jacoby, W. R., 1981. On modeling the lithosphere in mantle convection with non-linear rheology, *J. Geophys.*, **50,** 89–100.

Seama, N. & Isezaki, N., 1990. Sea-floor magnetization in the eastern part of the Japan Basin and its tectonic implications, *Tectonophysics*, **181,** 285–297.

Steiger, R. H. & Jäger, E., 1977. Subcommission on geochronology; convention on the use of decay constants in geo– and cosmo–chronology, *Earth planet. Sci. Lett.*, **36,** 359–362.

Tai, Y. & Kato, M., 1979. Iwami-Oda area, Shimane Prefecture, in *Fundamental Data on Japanese Neogene Bio- and Chrono-Stratigraphy*, pp. 101–102, IGCP-114 National Working Group of Japan.

Zhao, X., Coe, R. S., Zhou, Y. Wu, H. & Wan, J., 1990. New

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026

paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan, *Tectonophysics*, **181**, 43–81.

Zheng, Z., Kono, M., Tsunakawa, H., Kimura, G., Wei, Q., Zhu, X. & Hao, T., 1990. Apparent polar wander path for North China Block since Jurassic, *Geophys. J. Int.*, **104**, 29–40.

Zijderveld, J. D. A., 1967. A.C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism, Developments in Solid Earth Geophysics*, vol. 3, pp. 254–286, eds Collinson, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, Amsterdam.

Downloaded from https://academic.oup.com/gji/article/105/2/397/705554 by Curtin University Library user on 06 May 2026